IIT ROORKEE

NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-42

Dual Active Bridge Converter

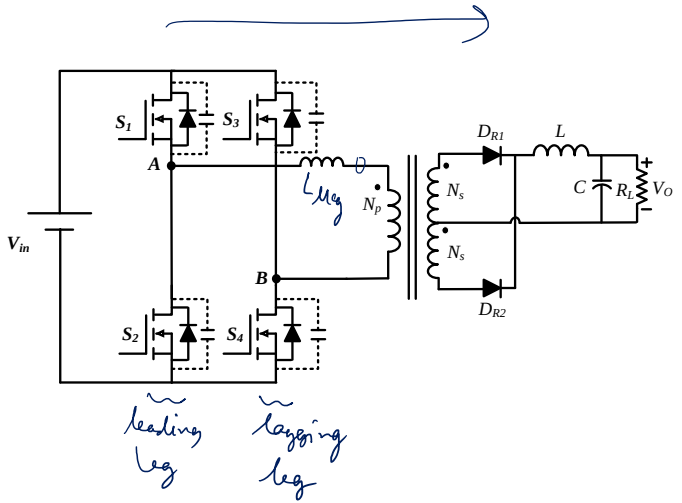
Dr. Apurv Kumar Yadav

Department of Electrical Engineering



Recap

- Full bridge converters: PWM schemes, PSFB
- Operating modes: ZVS turn-on conditions
- Small signal modelling

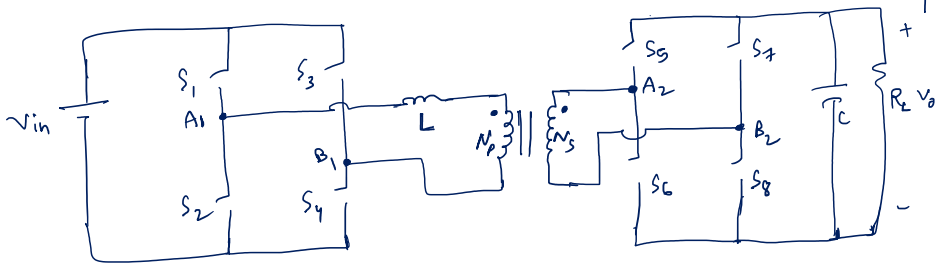


Disadvantages

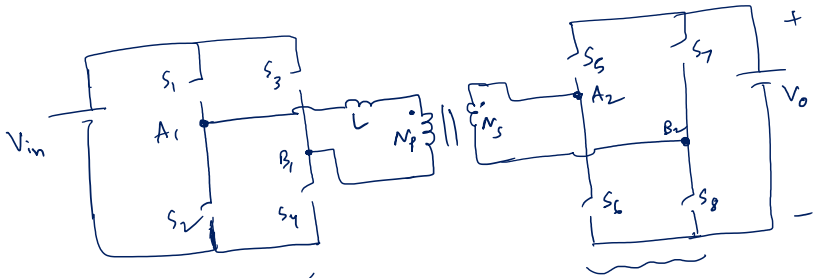
1. No bidirectional power flow
2. limited range of soft switching
3. large inductive filter in the secondary side
4. Duty walking (solved by peak current mode control)

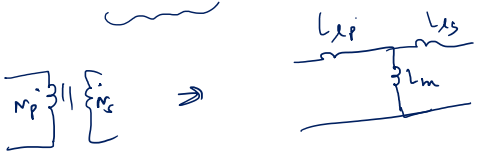
Dual Active Bridge (DAB)



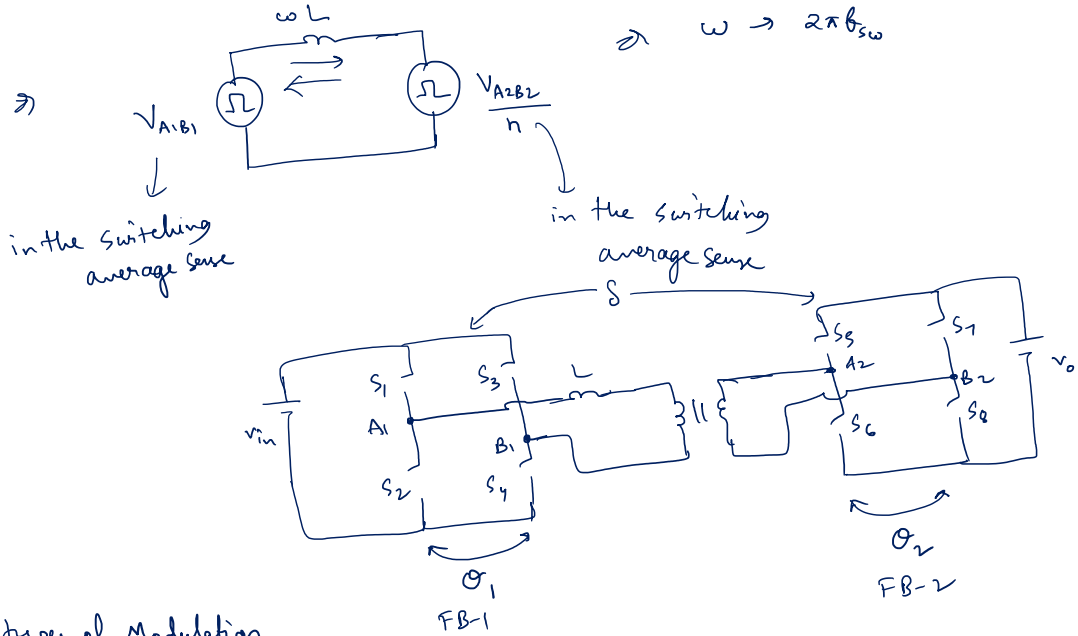


$\Rightarrow$





if L_{lp}, L_{ls} are small
L_m is large



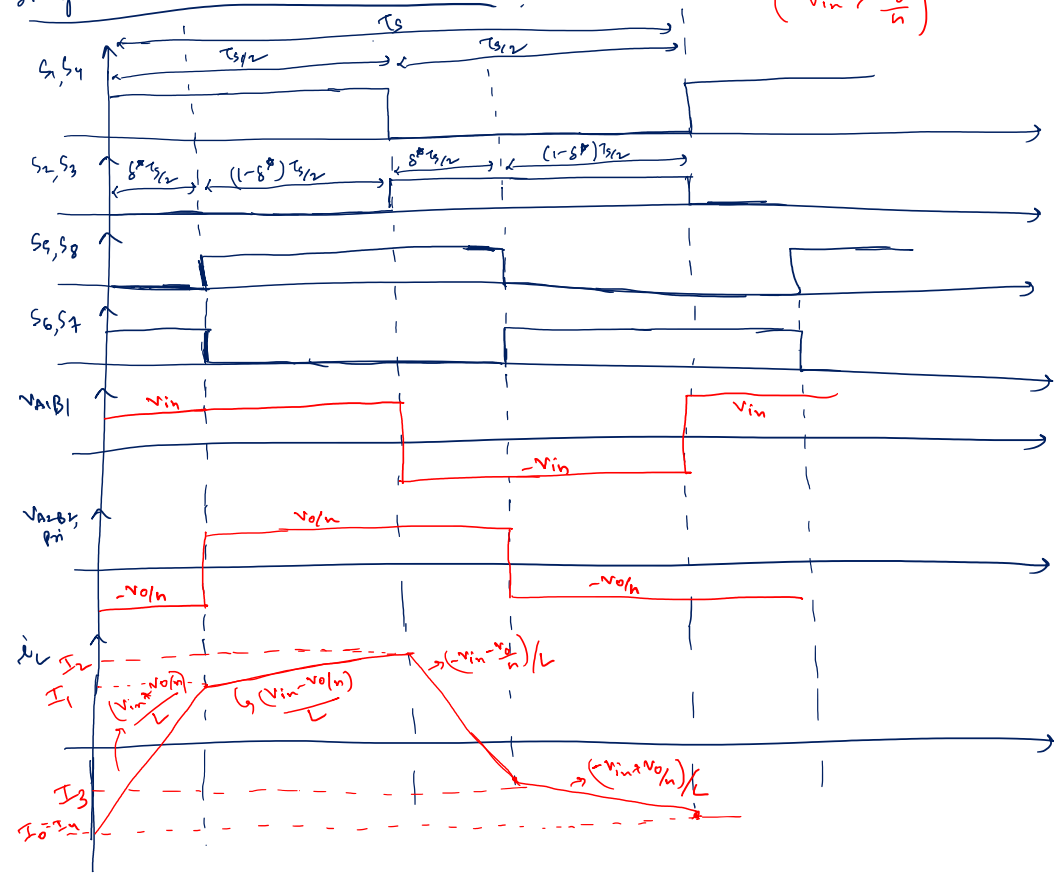
Different types of Modulation

1. Single phase shift (SPS) → 'δ'
2. Dual phase shift (DPS) → 'δ', θ_1 or 'δ', θ_2 or θ_1, θ_2
3. Triple phase shift (TPS) → $\delta, \theta_1, \theta_2$
4. Triple phase shift with variable frequency (TPSVF) → $\delta, \theta_1, \theta_2, f_{sw}$

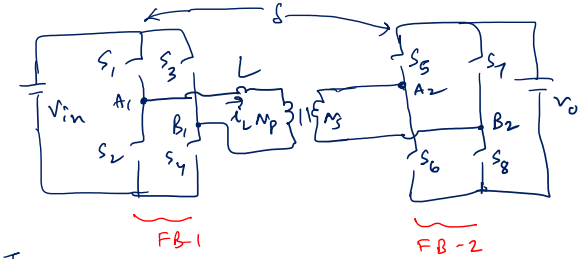
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4. Triple Phase Shift with variable frequency (TPSVF) → $\delta, \theta_1, \theta_2, f_{sw}$
5. Dual Phase Shift with variable frequency → very rarely used → f_{sw}, δ, θ_2 or f_{sw}, δ, θ_1 or $\theta_1, \theta_2, f_{sw}$

Single phase shift modulation



$(v_{in} > \frac{v_o}{n})$



$360^\circ \rightarrow T_s$

$180^\circ \rightarrow T_s/2$

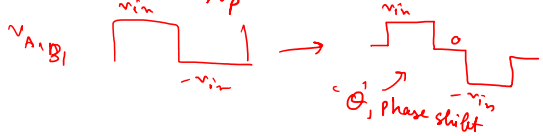
$\delta \rightarrow \frac{\delta}{180} \frac{T_s}{2}$

$\delta^* = \delta/180$

$\delta^* T_s/2$

$v_{A_2B_2, pri} \rightarrow v_{A_2B_2}$ referred to primary side

$n = \frac{N_s}{N_p}$



$\Rightarrow I_2 = -I_0$

$I_1 = -I_3$

$I_4 = I_0$

$\Downarrow$

This ensure no DC offset in the transformer primary current

Thank You